

1. Extend $\deg(0) = -\infty$.

$$\begin{aligned} f &= qg + r \\ f(t) &= f_d t^d + \cdots, f_d \neq 0 \\ q(t) &= a_n t^n + \cdots, a_n \neq 0 \\ g(t) &= b_m t^m + \cdots, b_m \neq 0 \end{aligned}$$

For existence:

$$\begin{aligned} P(n): \text{ for every } f \in F[t] \text{ with } \deg(f) = n, n \in \mathbb{N}, g \neq 0, \exists q, r: f &= qg + r, \deg(r) < \deg(g) \\ f = 0: 0 &= 0 \cdot g + 0 \\ \deg(0) &= -\infty < \deg(g) \end{aligned}$$

- base

$$\begin{aligned} P(n) \text{ is true for } n < \deg(g) \\ q = 0, r = f, f &= qg + r, \deg(r) = \deg(f) = n < \deg(g) \end{aligned}$$

- induction step, assume for every $n \leq N$: $P(n)$ is true, prove it for $N + 1$.

$$\begin{aligned} g_m \in F, F \text{ is a field, } g_m \neq 0 &\implies \exists g_m^{-1} \\ \text{let } a &= g_m^{-1} f_{N+1} \\ \text{let } f' = f - ag_m t^{N+1-m} &\implies \deg(f') < \deg(f) \implies f' = q'g + r \implies f = (q' + at^{N+1-m})g + r \end{aligned}$$

which concludes the induction.

For uniqueness assume that $\exists q', r': q \neq q' \text{ or } r \neq r', f = q'g + r'$.

$$\begin{aligned} f &= qg + r, f = q'g + r' \\ qg + r - (q'g + r') &= 0 \\ (q - q')g + (r - r') &= 0 \\ \deg(r) < \deg(g), \deg(r') < \deg(g) &\implies \deg(r - r') < \deg(g) \implies \deg(r' - r) < \deg(g) \end{aligned}$$

Consider two cases:

$$q = q' \implies r = r'$$

contradiction with initial assumption

$$\begin{aligned} q \neq q' &\implies \deg(q - q') \geq 0 \implies \deg((q - q')g) \geq \deg(g) \\ (q - q')g &= r' - r \implies \deg((q - q')g) = \deg(r' - r) \end{aligned}$$

which is a contradiction since $LHS \geq \deg(g), RHS < \deg(g)$, therefore $q' = q, r' = r$.

2. Let's use Euclidean algorithm for finding $\gcd(g, f)$:

- (a) if $f = 0$ result is g
- (b) otherwise result is $\gcd(r, g)$ where $f = qg + r$.

Such q, r exist (see problem 1), $\deg(r) < \deg(g)$. Therefore steps of Euclidean algorithm will look like:

$$\begin{aligned} f &= q_0g + r_0 \\ g &= q_1r_0 + r_1 \\ r_0 &= q_2r_1 + r_2 \\ &\dots \\ r_{k-1} &= q_{k+1}r_k + r_{k+1}, r_{k+1} = 0 \end{aligned}$$

Since $\deg(r_k)$ decreases with each step this algorithm is going to terminate in a result d .

Let's prove that $d \mid f, d \mid g$. To do that we will do reverse Euclidian algo steps to reconstruct f, g from d using existing coefficients q_i, r_i :

$$\begin{aligned} r_k &= d, r_{k+1} = 0 \\ d \mid r_i, d \mid r_{i+1}, r_{i-1} = q_{i+1}r_i + r_{i+1} &\implies d \mid r_{i-1} \\ \text{therefore } d \mid r_1, d \mid r_0 &\implies d \mid g, d \mid f \end{aligned}$$

Now let's consider an arbitrary c , such that $c \mid f, c \mid g$.

$$\begin{aligned} c \mid f, c \mid g &\implies c \mid r_0 \\ c \mid g, c \mid r_0 &\implies c \mid r_1 \\ &\dots \\ c \mid r_k, c \mid r_{k+1} &\implies c \mid r_{k+2} \end{aligned}$$

Since $\exists k: d = r_k$ then $c \mid d$ for arbitrary c . Therefore since $d \mid f, d \mid g, d = \gcd(f, g)$.

Now since d must be monic let's consider $d(t) = d_0 + d_1t + \dots + d_nt^n$. Since $d_n \in F, F$ — field $\implies \exists d_n^{-1} \implies d' = d_n^{-1}d$ is monic.

3. a) Assume $\gcd(\pi, f) = c$, then $c \mid \pi, c \mid f$. Since π is irreducible then $\pi = cd$, and either c or d is a unit. There are two possibilities:

- c is a unit. But then only monic unit is 1 and $\gcd(\pi, f) = 1$.
- d is a unit, c is an irreducible polynomial. But since π is irreducible then c and π are from the same equivalence class. But then $\gcd(\pi, f) = \pi$ and $\pi \mid f$ which is a contradiction.

- b) $\pi \nmid fg$. Assume $\pi \nmid f$ and $\pi \nmid g \implies \gcd(\pi, f) = 1$.

$$\begin{aligned} 1 &= \pi x + fy \\ g &= \pi gx + fgy \\ \pi \mid \pi gx, \pi \mid fgy &\implies \pi \mid g \text{ — contradiction} \end{aligned}$$

therefore $\pi \mid f$ or $\pi \mid g$.

- c) define the following algorithm. Let's say $fact(n)$ is a finite product of irreducible polynomials:

- if f is irreducible $fact(f) = f$
- if f is reducible $\exists h, g: f = hg \implies fact(f) = fact(h)fact(g)$.
- $\deg(h) < \deg(f), \deg(g) < \deg(f), \deg \in \mathbb{N}$ and $<$ on $\mathbb{N}$ is well-founded. $NSTEPS(call)$ — number of steps needed for termination.
 $P(n) = \forall f: \deg(f) = n, NSTEPS(fact(f)) < \infty$
 $P(1)$ is true since such f are irreducible.
Assume $P(n)$ is true for all $n \leq N$. Prove for $N + 1$.
For arbitrary $f: \deg(f) = N + 1, NSTEPS(fact(f)) = 1$ if it's irreducible or $NSTEPS(fact(f)) = NSTEPS(fact(g)) + NSTEPS(fact(h)) + 1$ otherwise $\implies P(N + 1) \text{ true}$.

d) consider arbitrary π_i . If $\pi_i \mid f$ then from 3.b there are two possibilities:

- $\exists \rho_j: \pi_i \mid \rho_j$. This means that ρ_j is either reducible which is a contradiction or $\rho_j = h\pi_i$ where $h \in F[t]^\times$, which means π_i and ρ_j are associates.
- $\pi_i \mid v$, but this is impossible because $\deg(v) < \deg(\pi_i)$.

To prove 1-to-1 matching of π_i and ρ_j and $r = s$ do the following:

- Remove u from the left hand side by multiplying both sides by u^{-1} and reassigning $v = vu^{-1}$. Since all π_i and ρ_j are monic then their products leading coefficients are equal to 1 which means $v = 1$ as well and can be removed.
- Suppose $\pi_1 \cdots \pi_r = \rho_1 \cdots \rho_s$. Then $\exists j: \pi_1 = \rho_j$.
 $F[t]$ is an integral domain $\implies$ multiplication is commutative $\implies$ we can reorder and get

$$\pi_1 \cdots \pi_r = \rho_j \rho_1 \cdots \rho_{j-1} \rho_{j+1} \cdots \rho_s$$

$F[t]$ is an integral domain $\implies$ 0 has no non-zero divisors which allows cancellation and reindexing:

$$\pi_2 \cdots \pi_r = \rho_2 \cdots \rho_s$$

Since number of factors is finite this process repeats until no more factors left on left or right side of the equality.

- Assume $r \neq s$, for example $r < s$ then after this process ends we get

$$1 = \rho_k \cdots \rho_s$$

Which is impossible because $\deg(1) = 0, \deg(\rho_k \cdots \rho_s) \geq 1$.

Similarly for $r > s$. Contradiction.